Chengkai Wu

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Education

The Hong Kong University of Science and Technology (Guangzhou) 2025/02 --
Ph.D Student in Robotics and Autonomous Systems Guangzhou, China

Harbin Institute of Technology, Shenzhen 2022/09 -- 2024/12
M.Eng in Control Engineering Shenzhen, China
- Outstanding Graduate

Xidian University 2018/09 -- 2022/06
B.Eng in Electronic Information Engineering Xi'an, China
- GPA: 3.8/4.0, Rank: Top 1%

Research Interests

My research interests center on robotics motion planning and control, utilizing both optimization and learning-based methods to generate safe and smooth locomotions.

Publications

Synergizing Efficiency and Reliability for Continuous Mobile Manipulation
Chengkai Wu, Ruilin Wang*, Yixin Zeng*, Jiayuan Wang, Mingjie Zhang, Guiyong Zheng, Qun Niu, Juepeng Zheng, Jun Ma, Boyu Zhou†.
In Submission. [Video]

Real-time Whole-body Motion Planning for Mobile Manipulators Using Environment-adaptive Search and Spatial-temporal Optimization Yokohama, Japan
Chengkai Wu*, Ruilin Wang*, Mianzhi Song, Fei Gao, Jie Mei, Boyu Zhou†.
2024 IEEE International Conference on Robotics and Automation (**ICRA 2024**). [Paper] [Video] [Code] - **Oral, Popular(14)**

FERMI: Flexible Radio Mapping with a Hybrid Propagation Model and Scalable Autonomous Data Collection
Yiming Luo, Yunfei Wang, Hongming Chen, **Chengkai Wu**, Ximin Lyu, Jinni Zhou, Jun Ma, Fu Zhang, Boyu Zhou†.
Robotics: Science and Systems 2025 (**RSS 2025**). [Paper] [Code]

ApexNav: An Adaptive Exploration Strategy for Zero-Shot Object Navigation with Target-centric Semantic Fusion
Mingjie Zhang, Yuheng Du, **Chengkai Wu**, Jinni Zhou, Zhenchao Qi, Jun Ma, Boyu Zhou†.
IEEE Robotics and Automation Letters 2025 (**RAL 2025**). [Paper] [Video] [Code]

Technical Skills

- **Programming Languages**: C/C++(ROS), Python, MATLAB
- **Tools**: Gazebo, Isaac Sim, Unity, SolidWorks, PX4, Git, LBFGS, ACADOS, Pytorch, LaTeX
- **Theoretical Knowledge**: Motion Planning, Numerical Optimization, Forward/Inverse Kinematics, Model Predictive Control (MPC), Reinforcement Learning
- **Languages**: Chinese (Native), English (IELTS: 7)

Services

- **Reviewer**: T-ASE, RA-L, ICRA
- **Teaching Assistant**: RBCC-Phase 3 (HKUST-GZ 2025 Summer), Calculus II (HKUST-GZ 2026 Spring)

Selected Honors and Awards

Outstanding Graduate - Harbin Institute of Technology 2025
Third Place - DJI RoboMaster 2023-2024 University AI Challenge - Classic Apr. 2024
Second Place - DJI RoboMaster 2022-2023 University AI Challenge Nov. 2022
School First-Class Academic Scholarship 2020,2023
First-class Senior Scholarship (2/495) Dec. 2020